Lec2:
Distributed Database

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Outline

- **Concepts of Distributed database systems.**
- **Distributed Data Processing.**
- **Layers of Transparency**

Concepts of Distributed database systems.

- Collection of logically-related shared data.
- Data split into fragments.
- Fragments may be replicated.
- Sites linked by a communications network.
- Data at each site is under control of a DBMS.

Distributed Data Processing

- **Distributed Databases Versus Distributed Processing**

The terms distributed database and distributed processing are closely related, yet have distinct meanings. There definitions are as follows:

- **Distributed database** **A set of databases in a distributed system that can appear to applications as a single data source.**
- **Distributed processing** **The operations that occurs when an application distributes its tasks among different computers in a network. For example, a database application typically distributes front-end presentation tasks to client computers and allows a back-end database server to manage shared access to a database.**

Promises of Distributed Databases

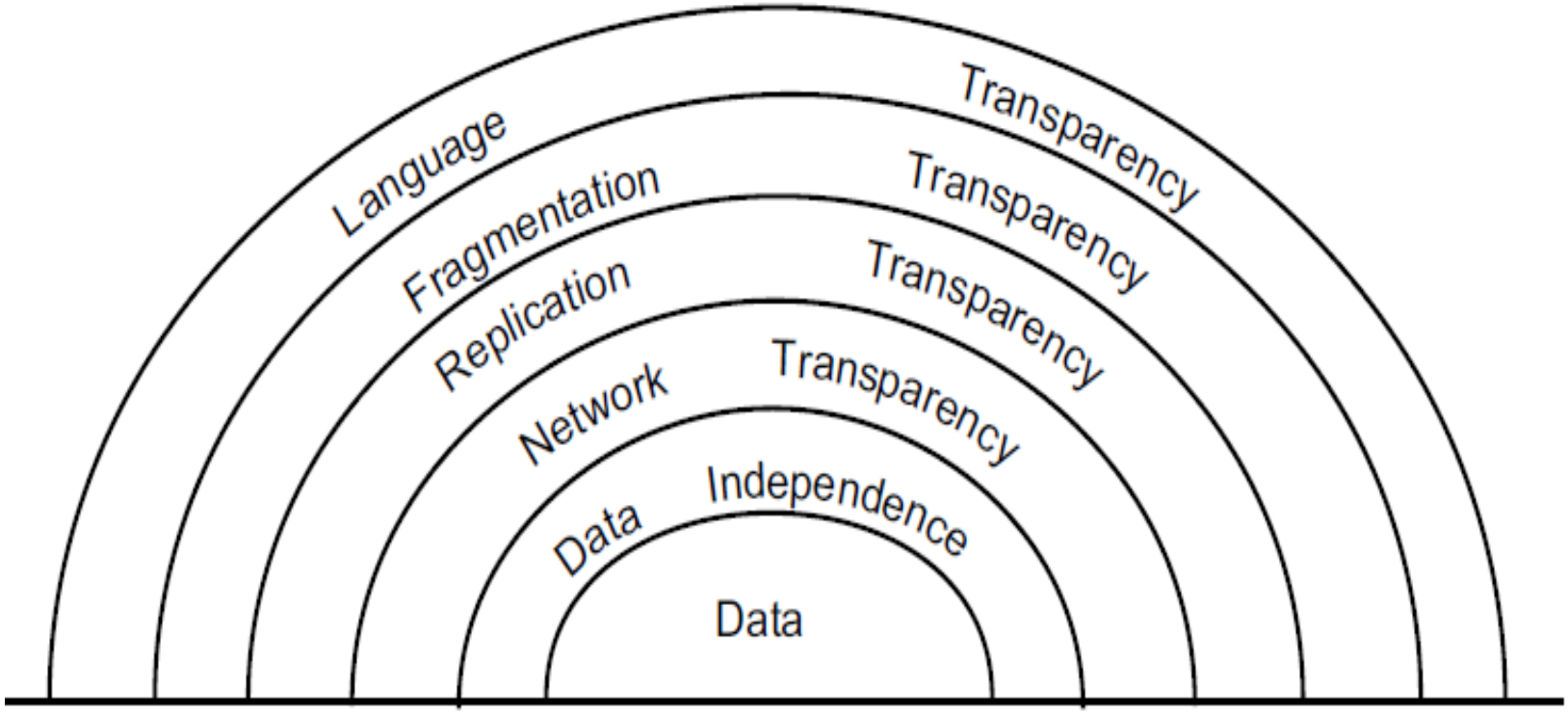
- Transparent management of distributed and replicated data.
- Reliable access to data through distributed transactions
- Improved performance.
- Easier system expansion.

Transparent Management of Distributed and Replicated Data

- Transparent system “hides” the implementation details from users.
- The advantage of a fully transparent DBMS is the high level of support that it provides for the development of complex applications

Layers of Transparency

- The system needs to be able to deal with a number of different types of transparencies.



Data Independence

- It refers to the immunity of user applications to changes in the definition and organization of data, and vice versa.
- **Types of data independence:**
  - Logical data independence
  - physical data independence..

Data Independence

- Logical data independence refers to the immunity of user applications to changes in the logical structure (i.e., schema) of the database.
- Physical data independence, on the other hand, deals with hiding the details of the storage structure from user applications

Network Transparency

- This type of transparency is referred to as network transparency or distribution transparency.
- The user should be protected from the operational details of the network; possibly even hiding the existence of the net.
- Distribution transparency requires that users do not have to specify where data are located.
- **Two types of distribution transparency :**
  - location transparency
  - naming transparency.

Network Transparency

- **Location transparency** refers to the fact that the command used to perform a task is independent of both the location of the data and the system on which an operation is carried out.
- **Naming transparency** means that a unique name is provided for each object in the database.

Replication Transparency

- The transparency issue is whether the users should be aware of the existence of copies or whether the system should handle the management of copies and the user should act as if there is a single copy of the data.
- Remember that replication transparency refers only to the existence of replicas, not to their actual location.

Fragmentation Transparency

- When database objects are fragmented, we have to deal with the problem of handling user queries that are specified on entire relations but have to be executed on subrelations. In other words, the issue is one of finding a query processing strategy based on the fragments rather than the relations.
- Typically, this requires a translation from what is called a global query to several fragment queries.

Who Should Provide Transparency?

- It is possible to identify three distinct layers at which the transparency services can be provided.
- the access layer.
- the operating system level
- The third layer at which transparency can be supported is within the DBMS.

Question?